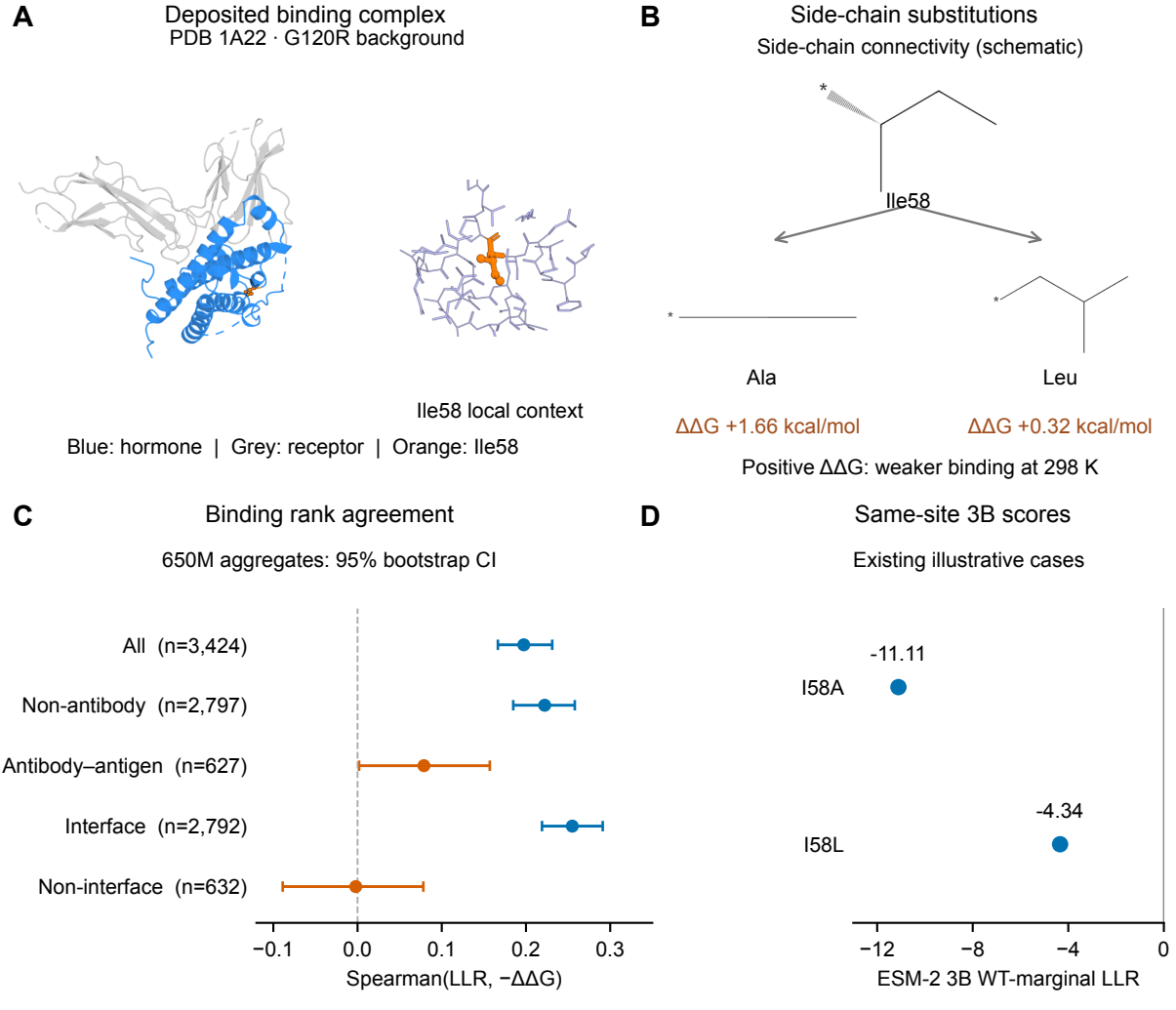

A
Deposited binding complex
PDB 1A22 · G120R background
Ile58 local context
Blue: hormone | Grey: receptor | Orange: Ile58
B
Side-chain substitutions
Side-chain connectivity (schematic)
Ile58
Ala
Leu
ΔΔG +1.66 kcal/mol
ΔΔG +0.32 kcal/mol
Positive ΔΔG: weaker binding at 298 K
C
Binding rank agreement
650M aggregates: 95% bootstrap CI
All (n=3,424)
Non-antibody (n=2,797)
Antibody–antigen (n=627)
Interface (n=2,792)
Non-interface (n=632)
−0.1
0.0
0.1
0.2
0.3
Spearman(LLR, −ΔΔG)
D
Same-site 3B scores
Existing illustrative cases
-11.11
I58A
-4.34
I58L
−12
−8
−4
0
ESM-2 3B WT-marginal LLR